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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S, FE) / S2 (PT) (S) Examination January 2024 (2019 Scheme)

Course Code: EET204

Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

		Marks
1	Explain the properties of divergence of vector field	3
2	Convert the point P(1,3,5) from Cartesian to cylindrical and spherical co-ordinate system	3
3	Describe about various types of charge distributions in Electric field	3
4	A point charge Q=0.4nC is located at the origin. Obtain the absolute potential at point A (2,2,3).	3
5	Find the current density for the magnetic field intensity $\overline{H} = r^2 sin\theta \bar{a}_{\varphi}$.	3
6	Find the magnetic field intensity at a point located on Y-axis due to an infinitely long current carrying element lying on Z-axis using Ampere's Circuital law.	3
7	What is intrinsic impedance of Electromagnetic waves	3
8	Explain the significance of skin depth in Electromagnetic field	3
9	What are the causes of Electromagnetic interference.	3
10	Define the characteristic impedance of a lossless transmission line.	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Explain how a point can be expressed in Spherical co-ordinate system.	4
	b)	Verify Stoke's theorem for a vector field $\overline{F} = r^2 cos\emptyset \, \bar{a}_r + z \, sin\emptyset \, \bar{a}_z$ around the path defined by $0 \le r \le 3, \quad 0 \le \emptyset \le 45^o$ and $Z = 0$.	10
12	a)	Transform the vector $\overline{B} = z\bar{a}_x + (1-x)\bar{a}_y + \frac{y}{x}\bar{a}_z$ into cylindrical coordinates.	8
	b)	Explain the physical interpretation of gradient and curl of a vector field.	6

Module -2

13	a)	Derive the expression of electric field intensity at $z = r$ due to an infinite line of	10

charge in XY plane extending from $-\infty$ to ∞ in both directions and having a uniform charge density of ρ_S C/m^2.

b) Find the electric flux density and volume charge density if the electric field intensity $\bar{E} = xy\bar{a}_x + x^2\bar{a}_y$. 4

14 a) Explain Poisson's and Laplace Equation in Electric field 4

b) A charge configuration in cylindrical co-ordinates is given by $\rho = 5re^{-2r}C/m^3$. Using gauss's law find electric flux density. 10

Module -3

15 a) A homogeneous dielectric ($\mu_r = 15$) fills region 1 ($z < 0$), while region 2 ($z > 0$) is filled by dielectric with $\mu_r = 1$. If $\bar{B}_1 = 1.2\bar{a}_x + 0.8\bar{a}_y + 0.4\bar{a}_z\ T$. Determine $\bar{B}_2$ and $\bar{H}_2$ in other medium and also calculate the angles made by the fields with the normal. 10

b) Compare conduction current and displacement current densities 4

16 a) A vector magnetic potential $\bar{A} = (2x^2y + yz)\bar{a}_x + (x^2y - x^3z)\bar{a}_y + (-6xy + 4y^2z^2)\bar{a}_z$ Wb/m. Find the magnetic lux through a loop described by $x = 1, y < 2, z < 2$. 10

b) Derive the differential and integral form of Maxwell's equation from Faraday's law. 4

Module -4

17 a) A plane wave in a nonmagnetic medium has $\bar{E} = 50\sin(10^8t + 2z)\,\bar{a}_y\ V/m$. Find i) the direction of wave propagation ii) wavelength iii) frequency iv) relative permittivity of the medium v) $\overline{H}$. 10

b) Define Phase velocity and group velocity. 4

18 a) Derive Poynting theorem for conservation of energy in an electromagnetic field and discuss the physical significance of each term in resulting equation. 10

b) A plane wave is propagating through polystyrene. Calculate the phase constant and velocity of propagation of wave if the frequency is 7000MHz. Given $\epsilon_r = 2.56$. 4

Module -5

19 a) What is impedance matching? Describe any two impedance matching methods used. 6

b) A transmission line has the following constants measured per km, Resistance = 10.15Ω, Inductance 3.93mH, Capacitance 0.00797μF Conductance 0.29μmho. 8

Determine the characteristics impedance and propagation constant at a frequency of 796Hz.

20 a) Define attenuation constant and phase constant. Derive the expressions of these constants in terms of primary constants of a line. 10

b) A 30m-long lossless transmission line with characteristic impedance 50 ohm operating at frequency of 2 MHz is terminated with a load of 60 + j40 ohm. Find reflection coefficient and standing wave ratio of the line. 4

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